

HAOMING WANG

PH.D. CANDIDATE

Presenting CVPR'26 *Oral* – InfiniBench – in Denver, June 5–7. Happy to meet at Booth 429.

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RESEARCH INTERESTS

On-Device Multimodal AI, Computer Vision, Vision-Language Models, Robotics & Autonomous Mobile Systems, Generative-Model Interpretability.

EDUCATION

Sept 2022 – May 2027 (Anticipated)	Ph.D. in Electrical and Computer Engineering University of Pittsburgh, Pittsburgh, PA, USA Advisor: Prof. Wei Gao, Intelligent System Lab	
Sept 2018 – May 2022	B.Eng. in Automation Zhejiang University, Hangzhou, China Chu Kochen Honors College	GPA: 3.8/4.0

EXPERIENCE

Sept 2022 – Present	Graduate Student Researcher Intelligent System Lab, University of Pittsburgh, Pittsburgh, PA, USA
	• (May 2025 – Present) Led on-device VLM and computer-vision research: photo-realistic 3D benchmark synthesis (CVPR'26 Oral); on-device cross-frame reasoning deployed on Jetson Orin / Meta AR Glass / smartphone; co-developed VLM-based autonomous mobile systems on quadruped and wheeled robots. • (Nov 2024 – May 2025) Drove on-device LLM personalization (MobiSys'25 Xpert) and training-free fine-grained interpretability for diffusion / GAN / autoregressive image generators (FineXL). • (Sept 2022 – Oct 2024) Designed federated learning systems for heterogeneous AIoT fleets: gradient correction under data / device heterogeneity (MobiCom'25); federated learning with unbounded device staleness (AAAI'25).

PUBLICATIONS (PEER-REVIEWED)

1. **[CVPR'26 Oral] InfiniBench: Infinite Benchmarking for Visual Spatial Reasoning with Customizable Scene Complexity**
Haoming Wang, Qiyao Xue, Wei Gao.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 (Acceptance 25.4%, Oral).
[\[arXiv\]](#) [\[Code\]](#) [\[Data\]](#)
 - **Architected** a customizable framework for analyzing VLM spatial reasoning that generates an unlimited supply of photo-realistic 3D test scenes with user-controllable axes for object count, clutter, occlusion, and viewpoint.
2. **[MobiSys'25] Never Start from Scratch: Expediting On-Device LLM Personalization via Explainable Model Selection**
Haoming Wang, Boyuan Yang, Xiangyu Yin, Wei Gao.
Proceedings of the 23rd ACM International Conference on Mobile Systems, Applications and Services (MobiSys), 2025 (Acceptance 18.0%). [\[Paper\]](#)
 - **Developed Xpert**, an explainable model-selection method that warm-starts on-device LLM personalization from the closest pre-personalized checkpoint of a similar past user, eliminating cold-start cost on edge hardware.

3. **[MobiCom'25] When Device Delays Meet Data Heterogeneity in Federated AIoT Applications**
Haoming Wang, Wei Gao.
Proceedings of the 31st ACM International Conference on Mobile Computing and Networking (MobiCom), 2025 (Acceptance 17.1%). [\[Paper\]](#)
 - **Engineered** a federated AIoT training method that jointly disentangles and corrects for data heterogeneity and device delays via targeted gradient correction.
4. **[AAAI'25] Tackling Intertwined Data and Device Heterogeneities in Federated Learning with Unlimited Staleness**
Haoming Wang, Wei Gao.
Proceedings of the 39th AAAI Conference on Artificial Intelligence, 2025 (Acceptance 23.4%). [\[arXiv\]](#)
 - **Designed** a gradient-inversion technique that recovers usable training signal from arbitrarily stale device updates, enabling FL to converge on real heterogeneous mobile / IoT fleets.

PREPRINTS & PAPERS UNDER REVIEW

1. **MosaicThinker: On-Device Visual Spatial Reasoning for Embodied AI via Iterative Construction of Space Representation**
Haoming Wang, Qiyao Xue, Weichen Liu, Wei Gao.
Under Review, 2026. [\[arXiv\]](#)
 - **Built** a training-free pipeline that stitches multi-view RGB frames into a compact semantic map, boosting small on-device VLMs' cross-frame spatial reasoning by up to 40%; deployed on Jetson Orin, Meta AR Glass, and smartphone.
2. **Uncovering and Shaping the Latent Representation of 3D Scene Topology in Vision-Language Models**
Haoming Wang, Wei Gao.
Preprint, 2026. [\[Paper\]](#) [\[Code\]](#)
 - **Discovered** a VLM latent subspace encoding 3D scene topology and **introduced** a Dirichlet-energy regularizer that improves real-world spatial reasoning by up to 12.1% over standard SFT.
3. **HEVEN: Small Object Search and Navigation on VLM-Based Autonomous Mobile Systems**
Anonymous Authors (incl. Haoming Wang, 3rd author).
Double-Blind Under Review, 2026.
 - **Co-architected** the first VLM-based mobile system for small-object search, with a hierarchical spatial semantic tree and Tree-of-Thought planner; achieved 73.86% success rate ($3\times$ over SOTA) on a real quadruped robot.
4. **GlobalNav: Daily Object Navigation in VLM-Based Autonomous Mobile Systems with Aligned Local and Global Views**
Anonymous Authors (incl. Haoming Wang, 3rd author).
Double-Blind Under Review, 2026.
 - **Co-developed** an infrastructure-assisted VLM navigation system aligning ceiling-camera global views with the robot's first-person view; lifted daily-object navigation success from $<30\%$ to 90.91% in simulation, $\sim 85\%$ on a real wheeled robot.
5. **Deciphering Personalization: Towards Fine-Grained Explainability in Natural Language for Personalized Image Generation Models (FineXL)**
Haoming Wang, Wei Gao.
Under Review, 2025. [\[Paper\]](#)
 - **Architected FineXL**, a training-free method decomposing a personalized image generator's divergence from its base into orthogonal natural-language concepts with quantitative scores; cut explanation error by 56%.
6. **FreezeAsGuard: Mitigating Illegal Adaptation of Diffusion Models via Selective Tensor Freezing**
Kai Huang, Haoming Wang (co-author), Wei Gao.
Preprint, 2024. [\[Paper\]](#) [\[Code\]](#)
 - **Co-developed** a bilevel-optimization defense that learns which diffusion-model tensors to lock so users

cannot fine-tune toward illegal content; 37–47% stronger mitigation than unlearning baselines.

7. Reasoning Path and Latent State Analysis for Multi-view Visual Spatial Reasoning: A Cognitive Science Perspective

Qiyao Xue, **Haoming Wang** (co-author), Weichen Liu, Shiqi Wang, Yuyang Wu, Wei Gao.

Under Review, 2026. [\[arXiv\]](#)

- **Co-authored** ReMindView-Bench, a cognitively grounded multi-view spatial-reasoning benchmark (50K+ VQAs across 15 VLMs).

HONORS & AWARDS

April 2026

ECE Research Assistant of the Year

Department of Electrical and Computer Engineering, University of Pittsburgh

TECHNICAL SKILLS

On-Device / Edge: Jetson Orin (PyTorch + TensorRT), Meta AR Glass, Android (ONNX / NNAPI), Unitree Go2, RealSense D435i.

Models & Tools: Stable Diffusion, CLIP, Gemini, GPT-4o/5, InternVL, Qwen-VL, LLaVA-Video, Grounding-SAM, DepthPro.

Programming: Python, PyTorch, CUDA, C/C++.